

A 2-primary Smith condition for maximal binary rank of skew walk matrices

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Abstract

Let S be an integer skew-symmetric matrix of order $n \geq 1$, let e be the all-ones vector, and suppose that the walk matrix $W = [e, Se, \dots, S^{n-1}e]$ is nonsingular. We prove that if its largest Smith invariant factor $d_n(W)$ is not divisible by 4, then $\text{rank}_{\mathbb{F}_2} W = \lceil n/2 \rceil$. This gives the square-free implication for every controllable oriented graph. The proof localizes the walk lattice at 2, obtains a cyclic quotient over $\mathbb{F}_2[t]$, and rules out a nonzero cyclic quotient carrying two compatible alternating forms. The argument is deductive and self-contained; historical novelty remains unverified.

1 Statement and context

Throughout, $e = (1, \dots, 1)^T$. For an integer skew-symmetric matrix S of order $n \geq 1$, define

$$W = W(S) = [e, Se, \dots, S^{n-1}e].$$

When $\det W \neq 0$, its positive Smith invariant factors are the integers $d_1 \mid d_2 \mid \dots \mid d_n$ for which there exist unimodular integer matrices P, Q with $PWQ = \text{diag}(d_1, \dots, d_n)$. The notation $\text{rank}_{\mathbb{F}_2} W$ means the rank of the reduction of W modulo 2.

Theorem 1.1. *For every integer $n \geq 1$ and every integer skew-symmetric matrix S of order n , if $\det W(S) \neq 0$ and $4 \nmid d_n(W(S))$, then*

$$\text{rank}_{\mathbb{F}_2} W(S) = \left\lceil \frac{n}{2} \right\rceil.$$

An oriented graph here is obtained by assigning one direction to each edge of a finite simple graph. Its skew-adjacency matrix has $s_{ij} = 1$ for an arc $i \rightarrow j$, $s_{ij} = -1$ for an arc $j \rightarrow i$, and $s_{ij} = 0$ otherwise. It is called controllable when $W(S)$ is nonsingular.

Corollary 1.2. *Let Σ be a controllable oriented graph on $n \geq 1$ vertices. If $d_n(W(\Sigma))$ is square-free, then $\text{rank}_{\mathbb{F}_2} W(\Sigma) = \lceil n/2 \rceil$.*

Proof. The skew-adjacency matrix is integer skew-symmetric, and a square-free integer is not divisible by 4. Apply Theorem 1.1. $\square$

The motivating question is Problem 1 in Section 5 (p. 15) of [1], which asks for the implication in Corollary 1.2. A saved copy of the first arXiv version confirms this attribution. No comprehensive comparison with the prior literature has been completed, and no claim of priority or historical novelty is made. Relative to that statement, Theorem 1.1 allows arbitrary integer skew-symmetric entries and replaces square-freeness by a condition only at the prime 2. As adjacent context on matrix rank over different coefficient fields and exact certification, we cite Zeng's work on Hadamard factorization [2]; no result from it is required below.

2 Two algebraic lemmas

If an endomorphism T acts on a vector space over $\mathbb{F}_2$, we view that space as an $\mathbb{F}_2[t]$ -module with t acting as T . Such a module is *cyclic* if some vector and its successive images under T span the space. The zero module is included in this convention. Every submodule of a cyclic $\mathbb{F}_2[t]$ -module is cyclic: writing the original module as $\mathbb{F}_2[t]/I$, the inverse image of the submodule is an ideal $J \supseteq I$. Since $\mathbb{F}_2[t]$ is a principal ideal domain, a generator of J generates J/I .

Lemma 2.1. *Under the hypotheses of Theorem 1.1, put $A = S \bmod 2$, $V = \mathbb{F}_2^n$, and $U = \text{im}(W \bmod 2)$. Then V/U is a cyclic $\mathbb{F}_2[t]$ -module, with t acting as A .*

Proof. Let

$$R = \mathbb{Z}_{(2)} = \{a/b \in \mathbb{Q} : b \text{ is odd}\}, \quad M = R^n, \quad L = WR^n.$$

The Cayley–Hamilton theorem gives $L = R[S]e$ and $SL \subseteq L$. Indeed, the first n powers of S applied to e generate L , and the monic characteristic polynomial expresses every later power in terms of these. In particular, $L/2L$ is a cyclic $\mathbb{F}_2[t]$ -module generated by $e + 2L$.

The integer Smith form remains a valid change of basis over R . Since $d_i \mid d_n$ and $4 \nmid d_n$, every d_i is a unit or twice a unit in R . Consequently

$$2M \subseteq L. \tag{1}$$

Thus M/L is annihilated by 2, and

$$\phi : M/L \longrightarrow L/2L, \quad \phi(x + L) = 2x + 2L$$

is a well-defined $\mathbb{F}_2[t]$ -linear map. If $2x \in 2L$, write $2x = 2\ell$ with $\ell \in L$. Cancellation of 2 in the torsion-free module M gives $x = \ell$, proving injectivity. It follows from the submodule fact above that M/L is cyclic. Finally, $M/2M$ identifies with V , and the image of L in V is U . By (1),

$$V/U \simeq M/L$$

as $\mathbb{F}_2[t]$ -modules, proving the assertion. $\square$

Lemma 2.2. *Let H be a finite-dimensional $\mathbb{F}_2$ -vector space with a nondegenerate alternating bilinear form b . Suppose an endomorphism T satisfies*

$$b(Tx, y) = b(x, Ty), \quad b(x, Tx) = 0 \quad (x, y \in H).$$

If H is cyclic for T , then $H = 0$.

Proof. Choose a cyclic vector z . Self-adjointness gives, for $i, j \geq 0$,

$$b(T^i z, T^j z) = b(z, T^{i+j} z).$$

When $i + j = 2k$, this equals $b(T^k z, T^k z) = 0$ because b is alternating. When $i + j = 2k + 1$, it equals $b(T^k z, T(T^k z)) = 0$ by the second hypothesis. The orbit of z spans H , so b vanishes identically. Nondegeneracy forces $H = 0$. $\square$

The condition $b(x, Tx) = 0$ is a separate hypothesis in Lemma 2.2; self-adjointness alone in characteristic 2 does not justify the odd-power step of the proof.

3 Proof of the rank identity

Retain A, V, U from Lemma 2.1 and write $r = \dim U$. The space U is the span of $A^j e$ for $j \geq 0$, again by Cayley–Hamilton. On V use the nondegenerate dot product $b(x, y) = x^T y$. Integer skew-symmetry of S implies that A is symmetric with zero diagonal. Hence

$$b(Ax, y) = b(x, Ay), \quad b(x, Ax) = 0 \quad (x, y \in V). \quad (2)$$

The second identity follows by pairing the off-diagonal terms in $x^T Ax$.

The lower bound

Both U and its orthogonal complement $U^\perp$ are A -invariant. Set

$$K = U \cap U^\perp, \quad H = U^\perp / K.$$

The radical of the restriction of b to $U^\perp$ is exactly K , because $(U^\perp)^\perp = U$. Thus b induces a nondegenerate bilinear form $\bar{b}$ on H . It is alternating: for $x \in U^\perp$, the inclusion $e \in U$ gives

$$b(x, x) = \sum_i x_i^2 = \sum_i x_i = b(e, x) = 0.$$

The operator A induces an endomorphism T on H , and the two identities in (2) descend to $\bar{b}$ and T .

Inclusion into V induces an injective $\mathbb{F}_2[t]$ -module map

$$H \longrightarrow V/U, \quad x + K \longmapsto x + U.$$

Its kernel is zero by the definition of K . Lemma 2.1 and the cyclic-submodule property show that H is cyclic. Lemma 2.2 now gives $H = 0$. Therefore $U^\perp \subseteq U$, and

$$n - r = \dim U^\perp \leq \dim U = r, \quad r \geq \left\lceil \frac{n}{2} \right\rceil. \quad (3)$$

The upper bound

This bound uses only that A is symmetric with zero diagonal. Put $m_k = e^T A^k e \in \mathbb{F}_2$ for $k \geq 0$. For odd $k = 2j + 1$,

$$m_{2j+1} = (A^j e)^T A(A^j e) = 0.$$

For $k = 2j \geq 2$, the characteristic-2 identity $x^T x = e^T x$ gives

$$m_{2j} = (A^j e)^T (A^j e) = e^T A^j e = m_j.$$

Repeated halving therefore proves $m_k = 0$ for every $k \geq 1$. Symmetry also gives

$$b(A^i e, A^j e) = m_{i+j} \quad (i, j \geq 0). \quad (4)$$

If n is even, $m_0 = 0$ as well. Equation (4) then shows that U is totally isotropic, meaning that b vanishes on $U \times U$. Thus $U \subseteq U^\perp$, so $2r \leq n$.

If n is odd, $m_0 = 1$. Let $U_+ = \text{span}\{A^j e : j \geq 1\}$. The moment identities show that $U_+ \subseteq e^\perp$ and that U_+ is totally isotropic. Since $b(e, e) = 1$, we have $U = \text{span}\{e\} \oplus U_+$. Moreover, the form restricted to $e^\perp$ is nondegenerate: its radical is $e^\perp \cap \text{span}\{e\} = 0$. The isotropic dimension bound in this $(n - 1)$ -dimensional space gives

$$2 \dim U_+ \leq n - 1, \quad r = 1 + \dim U_+ \leq \frac{n + 1}{2}.$$

In either parity, $r \leq \lceil n/2 \rceil$. Together with (3), this proves Theorem 1.1. For $n = 1$, the definitions give $S = [0]$ and $W = [1]$, consistently with the argument. $\square$

4 Scope and assistance

The arithmetic hypothesis controls only the 2-primary part of the Smith form. No restrictions on the odd prime powers in d_n are needed. The result assumes nonsingular W and the all-ones input vector; no assertion for arbitrary input vectors or converse is made. The proof establishes the displayed rank implication without invoking any spectral-determination criterion. The comparison with prior literature remains limited as stated in Section 1.

Generative AI (OpenAI Codex) assisted with proof development, exact computation, a separate mathematical review pass, and manuscript preparation. That review was a separate model context, not external peer review. Supplementary Python checks independently recomputed the Smith factors and binary ranks of two saved order-six examples, using all square minors and rational inverse denominators. A separate exhaustive check covered all 1,099 symmetric zero-diagonal binary matrices of orders 1 through 5: in all 777 cases with cyclic V/U , the rank was $\lceil n/2 \rceil$. These finite checks are diagnostics; no computation is a premise of the proof, and no proof-assistant formalization is claimed.

References

- [1] Limeng Lin, Wei Wang, Wei Wang, and Hao Zhang. *A New Sufficient Condition for Oriented Graphs Determined by Their Generalized Skew Spectra*. arXiv:2609.03428v1, 3 September 2026. Problem 1, Section 5, p. 15.
- [2] Zijian Zeng. *A Rational Rank-(2, 2) Hadamard Factorization of a Proposed 4×4 Real Counterexample*. Zenodo preprint, September 2026. <https://doi.org/10.5281/zenodo.22242517>.